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Review Article

Enhancing anaerobic digestion Efficiency: A comprehensive review on innovative intensification technologies

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ABSTRACT

Anaerobic digestion (AD) is an established technology that plays a crucial role in breaking down the organic compounds and biomass during the sludge treatment processes. However, there are multiple challenges associated with the application of AD on different feedstocks and under various operational conditions. The AD process is highly sensitive to operational conditions (e.g., temperature and pH) with relatively slow reactions rates especially during the hydrolysis and methanogenesis stages. These limitations can significantly affect the performance of anaerobic digesters and the biogas production rate. Therefore, various intensification technologies were proposed and investigated in the literature to upgrade the biogas production and yield as well as enhancing the removal of organics and biomass during the sludge treatment processes. Although different review studies have examined some of these intensification technologies such as physical and chemical pretreatment techniques, limited studies have focused on reviewing the innovative intensification technologies, such as microbial electrolysis cells (MEC) and micro-aeration, in AD applications. Moreover, there are no systematic investigations that compared the performance, mechanisms, advantages, and challenges of these innovative technologies to draw strong conclusions about the applicability of each technology with different wastes, feedstocks, and operation conditions. In addition, the quantification of possible integration of these technologies with the current infrastructure and the technology readiness level were not well-investigated in literature. Therefore, in the current study, seven different innovative intensification technologies were reviewed including MEC-assisted AD, conductive functional materials, micro-aeration, anaerobic membrane bioreactors, hydrogen injection, IntensiCarb, and microbial hydrolysis process using *Caldicellulosiruptor bescii*. A detailed description of these technologies for increasing biogas yields was presented, with a special focus on the performance, reliability, efficiency gains, and applicability of each technology. The major insights of this review can serve as a reference for the potential intensification technologies that can be integrated with existing AD systems for enhanced biogas production and removal of organics and biomass.

1. Introduction

Anaerobic digestion (AD) is a mature technique for efficient bioenergy production during sludge treatment processes [1,2]. AD process involves breaking down organic compounds into biogas (e.g., methane and carbon dioxide) with a reduction in the concentration of total solids in sludge under anaerobic conditions [3]. Compared to the typical aerobic treatment processes, anaerobic digesters have multiple advantages

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as they produce less sludge with higher biogas generation, requiring less nutrients and energy demands [4,5]. AD process consists of four main steps including hydrolysis, acidogenesis, acetogenesis, and methanogenesis [6]. In the hydrolysis reaction, complex organics are disintegrated into amino acids, sugars, and fatty acids which are further transformed into volatile fatty acids (VFAs) (e.g., butyrate and propionate) through the acidogenesis reactions [7]. These produced VFAs are converted into acetate and hydrogen by acetogenesis reactions. The produced hydrogen and acetate are then consumed by methanogenic microorganisms to produce methane and biogas [8]. The hydrolysis, acidogenesis, and acetogenesis steps take place by bacterial populations while the methanogenesis is performed by strict anaerobic archaea [9].

Despite the effectiveness of AD in the sewage sludge treatment, it has some limitations that hinder its wide applications in the digestion of other wastes and feedstocks such as lignin [10], animal manure [11,12], and lipid-rich wastewater [13]. In addition, AD process is highly affected by the operation conditions including temperature, pH, carbon-to-nitrogen ratio (C/N), and hydraulic retention time (HRT) [8,14–16]. This is mainly because of the diversity of microbial communities involved in the digestion process which are sensitive to various operational conditions [17]. For example, methanogens are slow-growing microorganisms and highly influenced by the operating temperature where mesophilic temperatures (i.e., 25 – 40 °C) and long solids retention times (SRT) are required for efficient digestion process and high methane yield [18–20]. Other challenges are associated with the AD process itself such as insufficient disintegration of substrates, low biogas yield, and sludge destabilization especially for complex substrates (e.g., lignocellulosic material) [6]. In addition, AD process requires long start-up time, specific nutrient and trace metals, and long recovery [21–24]. AD process can also lead to the accumulation of different short chain VFAs, causing process instability inside the digesters [17,25]. Ultimately, the AD processes have slow hydrolysis and methanogenesis rates which represent the most rate-limiting reactions in the digestion process [26]. This magnifies the importance of extending the SRT to more than 15 days in the digesters [27].

The need for substantial improvement in the economic feasibility and applicability of AD facilities is generally linked to higher biogas production and yields which has attracted significant attention to the intensification techniques to overcome the limitations of AD process. To this end, different intensification technologies have been proposed and applied to enhance the degradation of different substrates in AD systems to overcome the previously mentioned challenges in the AD processes [23]. Intensification technologies refer to biological, chemical, and physical techniques that can enhance the performance of AD systems and increase the produced biogas, overcoming the main challenges associated with these systems. These technologies include pretreatment [28–30], co-digestion [31–33], additives [6,34], and configuration-based technologies [15,35]. The process intensification can be defined as enhancing the process performance significantly with certain effort and resources within the existing infrastructure of sludge treatment facilities [27]. Overall, intensification technologies can enrich the microbial populations and enhance their activity which leads to higher biogas yield and production rate [36–38]. They can also allow higher organic and biomass removal rates [7], decreasing the sludge production rate and the associated operational cost for sludge transportation and disposal.

Although some previous review studies in literature examined some of the intensification technologies in the AD applications such as additives [36,39,40], in-situ systems [5,7,41], and pretreatment techniques [42–44], these studies were mainly devoted to pretreatments and anaerobic co-digestion techniques. While less attention and investigations have been paid to examining the innovative intensification technologies, such as microbial electrolysis cells assisted AD (MEC-AD) [45,46], hydrogen injection [41], and micro-aeration [2], in the AD applications. In addition, there is a lack of comprehensive reviews that describe these innovative intensification technologies within the context of overall process efficiency. Moreover, there are no systematic studies that compared different innovative technologies based on their efficiency gains, advantages, suitability, and potential integration with existing infrastructure of wastewater treatment plants. In addition, there is a lack of determining the suitable intensification technology for efficient AD processes and upgrading biogas production according to waste types, feedstocks, available resources, and associated operational and capital costs. Additionally, the possible integration and interplay between different AD intensification technologies and their combined effects on the performance of digesters (e.g., enhancement in biogas production and yield) has not been thoroughly examined. Ultimately, to the best of the authors' knowledge, there are no published reviews that determined the technology readiness level (TRL) of these AD intensification technologies. This highlights the importance of comparing the performance, environmental, and economic aspects of the innovative technologies to confirm their applicability in full-scale sludge treatment facilities. Therefore, the current review aims to fill these gaps by providing an exhaustive analysis of the latest advancements in intensification technologies for anaerobic digesters. Unlike previous studies, this review systematically describes and compares various innovative techniques, offering a holistic perspective on their potential synergies and cumulative impact on biogas production as well as the challenges associated with the application of each technology in full-scale AD systems.

The main objective of this review is to discuss and compare the main mechanisms, performance, and efficiency gains of various innovative intensification technologies for increasing biogas production in AD systems. Also, the major advantages and challenges associated with each intensification technology are highlighted from different perspectives (e.g., practical and economical). In addition, this review investigates the possibility of integrating these innovative technologies with the existing infrastructure facilities to assess the feasibility of employing these technologies in full-scale sludge treatment facilities. Ultimately, the TRL is quantified for each of these innovative intensification technologies.

2. AD intensification using innovative technologies

The following sections discuss multiple innovative technologies for intensification applications that aim to improve one or more of the steps in the AD process. These technologies include MEC-AD, conductive functional additives, micro-aeration, anaerobic membrane bioreactors (AnMBRs), hydrogen injection, IntensiCarb, and microbial hydrolysis process (MHP) using *Caldicellulosiruptor bescii (C.bescii)*. For each technology, the main mechanisms, configurations, and governing factors on the performance of technology are explained. In addition, the main advantages and challenges associated with each technology are highlighted. The efficiency gains by each technology are also illustrated and shown to determine the influence of each technology on enhancing the biogas production.

2.1. Microbial electrolysis cells (MECs)

MEC is an innovative bio-electrochemical technology that can enhance the degradation of substrates and biogas production rate of the AD system. In the MEC, organic matter is oxidized by exoelectrogenic bacteria (e.g., *Geobacter sulfurreducens* and *Shewanella oneidensis*) at the anode where protons and electrons are released during this reaction [47,48]. These electrons are transferred to the cathode side through an external circuit assisted by a small, applied voltage (0.2 – 0.8 V) where hydrogen gas is produced when the electrons and protons are combined under anaerobic conditions [3,49,50]. This generated hydrogen is considered a source of energy and can be also used to increase the pH level, preventing the inhibition of methane-producing archaea [51]. Although the typical electrical input for water electrolysis ranges from 1.8 to 3.5 V, the applied voltage in the MEC systems (0.2 – 0.8 V) is much lower because the microbial oxidation of the organic substrates at the

anode replaces the water-splitting reactions [51,52]. Therefore, MEC can be an efficient approach to produce methane and hydrogen from wastewater with low sludge yields [53–55].

MECs can be coupled with AD systems in two configurations: direct immersion of electrodes in the anaerobic digesters or integration of the anaerobic digesters with external MECs (i.e., circulating the sludge in a loop between MEC and digester) (Fig. 1) [56]. The second configuration is preferable since it can be an easily retrofittable technique for existing anaerobic digesters, providing a high level of flexibility in construction and operation [17]. In addition, the implementation of recirculation loop between MEC and digester can buffer the influent acidity and mitigate the inhibition occurred by nitrogen and organic overloads [57,58]. This is mainly because of the dilution of acidic compounds (e.g., VFAs) and fostering a stable and resilient microbial community with higher degradation of complex compounds. However, this configuration requires larger footprint and higher capital cost.

In the MEC-AD systems, methane is produced through four routes and mechanisms [17,59] (Fig. 2). In the first route, the protons are reduced into hydrogen at the cathode side then hydrogen is utilized by hydrogenotrophic methanogens. The second route depends on the production of methane by direct consumption of electrons, protons, and carbon dioxide by electrotrophic methanogens. For the third route, fermentative bacteria produce hydrogen that is utilized by hydrogenotrophic methanogens while acetoclastic methanogenesis represents the fourth route for producing methane.

Multiple factors can affect the performance of MEC-AD systems such as the applied voltage [60–63], organic loading rate [64], SRT [65], temperature [17], surface area of electrodes [66], circuit conditions (i.e., open or closed) [3], and electron exchange rates and pathways [67]. For example, Gajaraj et al. (2017) demonstrated that low voltage (i.e., 0.3 V) increased the population of hydrogenotrophic methanogens by 17.2 times because the steady and slow production of hydrogen at the cathode enhanced the growth of hydrogenotrophic methanogens, while it had a negligible effect on the growth of acetoclastic methanogens [62]. Chen et al. (2023) observed that increasing the applied voltage from 1.3 to 1.6 V increased methane production from 12.70 to 57.02 % and VS removal from 37.99 to 52.72 % [60]. Joicy et al. (2022) assessed the effect of changing the applied voltage on methane yield where it was demonstrated that relatively high voltage (i.e., 0.9 V) enhanced methane yield by 37.3 % compared to low voltage (i.e., 0.3 V) which improved methane yield only by 27.2 % [63]. Kanellos et al. (2024) observed that when the organic loading rate increased from 1.7 to 2.9 g-COD/L/d, the biogas production increased from 30 % to 32 % while COD removal was enhanced from 12 to 19 % [64]. Asztalos and Kim (2015) elucidated that increasing SRT from 7 to 14 days can enhance the COD removal by at least 10 % according to the operating temperature [65].

The usage of MEC-AD systems has a lot of advantages in which energy recovery, microbial communities, and methane production and yield are significantly enhanced. For example, the polarization of electrodes in the MEC-AD systems promotes the direct interspecies electron transfer (DIET) and enhances the main enzymes of the microbial communities involved in the digestion processes, providing stable and diverse biological community for efficient methane production [63,68]. Moreover, the energy recovery from the electricity used in operating the MEC-AD systems can reach up to 42 % which is coupled with an increase in the methane yield (≅ 9 %) [62].

The efficiency gains from the application of MEC-AD systems in previous studies varied according to the sludge source, wastewater (WW) type, and operational conditions (e.g., applied voltage and bovine serum albumin (BSA)) (Fig. 3). Multiple studies in the literature showed a significant increase in the methane yield, biogas production rate, COD and VSS removal using the MEC-AD systems compared to the control reactors [65,69,70]. For example, Zhao et al. 2021 highlighted that the efficiency of methanogenesis was improved by 225 % when the BSA increased from 0.5 to 4 g/L [69]. It was also observed that the abundance of fermenters at the anode and *Methanobacterium* at the cathode increased by 46.7 and 74.3 %, respectively, highlighting that the methanogenesis pathway in the MEC-AD was switched from the acetoclastic to hydrogenotrophic pathway [69]. The MEC-AD technology also enhanced the bioconversion of coal to methane by 39.08 % compared to the typical AD system with a significant enhancement in DIET between multiple microbes [71]. In a previous lab-scale study, the produced energy from MEC-AD system was higher than the input energy by approximately four times while achieving a COD removal of 12 % [72].

The main challenge in the implementation of MEC-AD technology on full-scale applications is the capital cost associated with the construction, electrodes and membranes where the capital cost of MEC can reach up to 5500 Euro/m^3 [17,46]. Most of the electrodes used in MEC-AD technology are carbon-based materials such as graphite plate, graphite rod, carbon cloth, and carbon brush [50,73–76]. Some transition metals (e.g., nickel, copper, and cobalt) can be also used as cathodes in the MEC-AD systems due to their corrosion resistance and electrocatalytic properties [59]. It should be also noted that the conductivity and catalysis of electrodes play a critical role in the efficiency of MEC-AD technology since they can control the flow of electrons and the resistance of the MEC [77]. In addition, highly conductive electrodes can facilitate the DIET process between microbial communities, enhancing the methane production rate [50,59]. Another challenge in applying

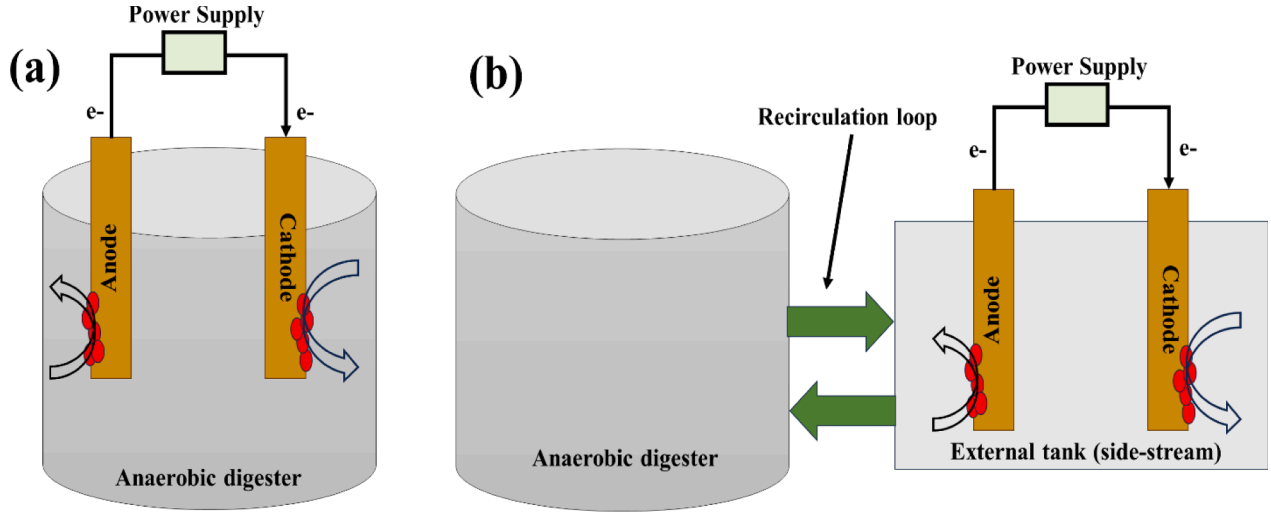


Fig. 1. MEC-assisted AD configurations: (a) immersed electrodes in the anaerobic digesters, and (b) integration between the anaerobic digesters and external MECs.

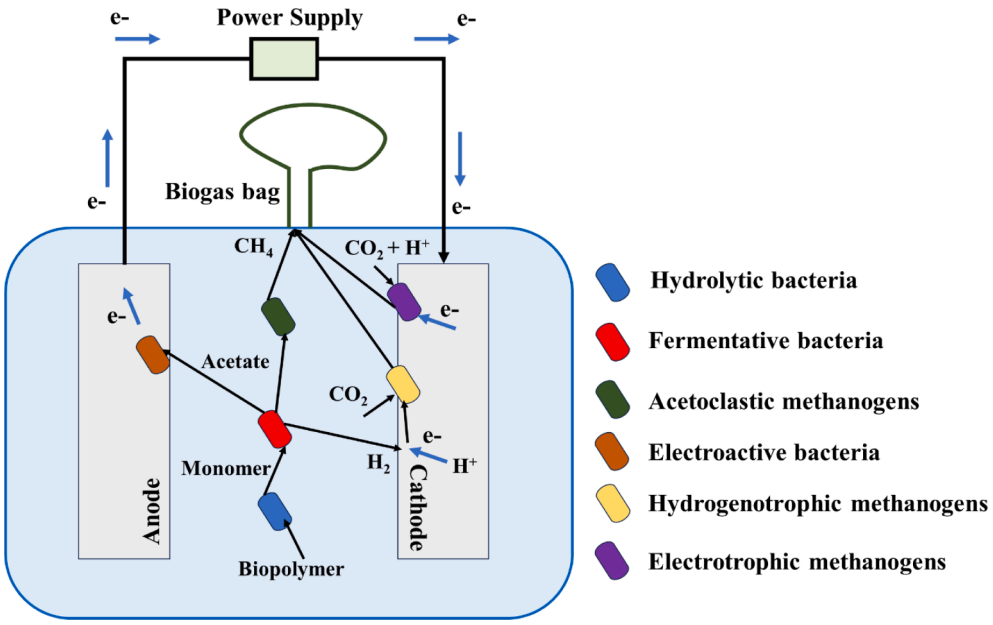


Fig. 2. Mechanisms of methane production and reaction pathways in the MEC-AD systems.

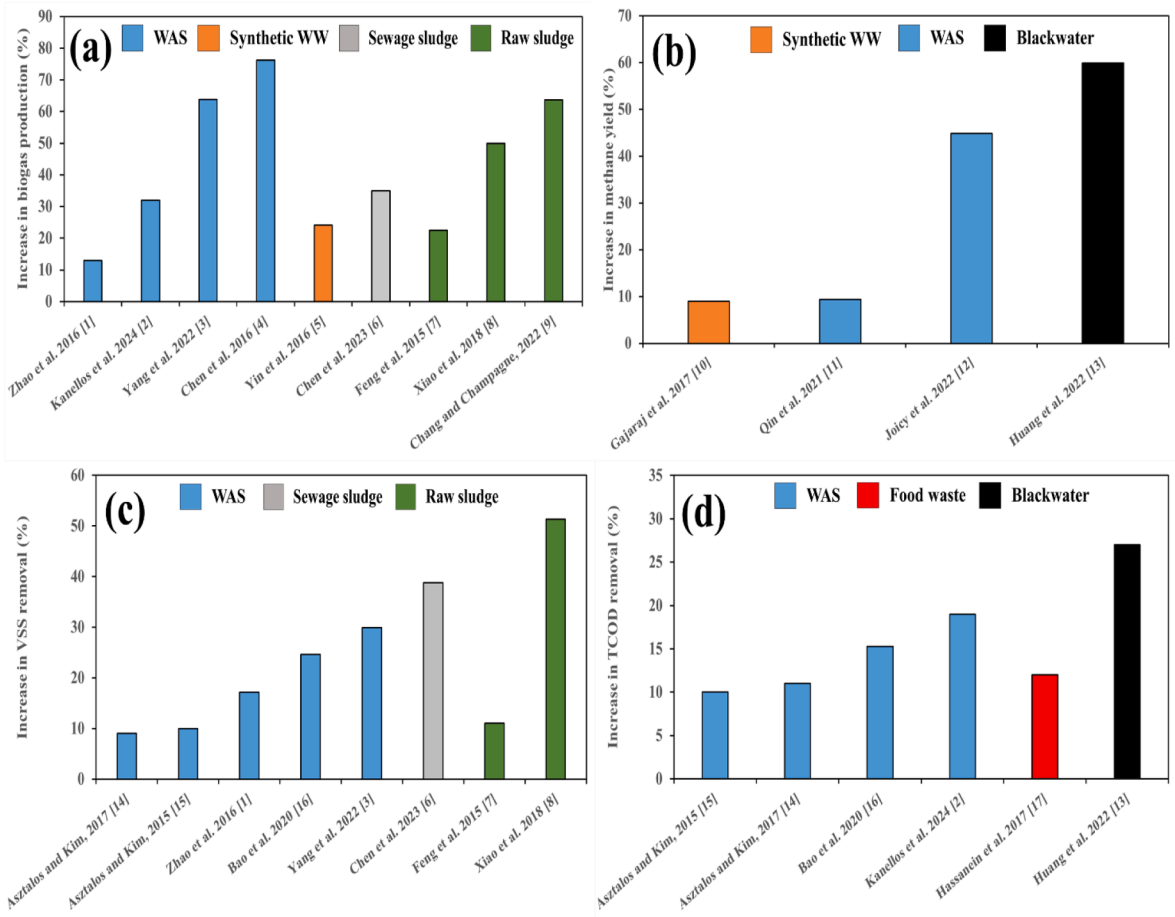


Fig. 3. Efficiency gains due to the application of MEC-assisted AD technology in previous studies through increasing: (a) biogas production, (b) methane yield, (c) VSS removal, and (d) TCOD removal. (All references used for preparing the figure are listed in the supplementary material section.).

MEC-AD technology in full-scale systems is the high operation cost associated with the electrical energy input required for MECs [17,46,78]. In addition, since the outcomes from previous MEC-AD studies were determined according to different operational conditions (e.g., applied voltage), it is extremely challenging to systematically formulate the interdependences between the various parameters

involved in the digestion process [17]. A major limitation of applying MEC-AD technology on full-scale facilities is the fouling of electrodes which can increase the resistance of mass transfer, deteriorating the system performance due to the accumulation of solids on electrodes especially at long-term operation [79,80].

2.2. Conductive functional materials

Conductive functional materials can be a practical technique to enhance the methane production in the anaerobic digesters. The main mechanism of conductive functional materials is that they replace the cytochromes and pilis of bacteria, working as an electron mediator between electrophilic methanogens and electroactive bacteria [81] (Fig. 4). There are various types of conductive additives including carbon-based materials (e.g., granular activated carbon (GAC), graphite, carbon cloth, and biochar), magnetite, and other iron-based additives that can be incorporated in the AD applications to enhance the biogas production and removal of VSS and COD [82]. One or more than one conductive additive can be applied in the anerobic digestion process (Fig. 5). For example, multiple previous studies used GAC as a conductive material for better digestion of food waste [83], swine manure [84], and blackwater [85]. While other studies deployed biochar as an effective alternative for high biogas production rate and methane yield in the digestion of sewage sludge [86,87], cow manure [11], and swine manure [12]. Different materials can be added together such as GAC and magnetite, which were added to allow better digestion of fat, oil, and grease, and food waste [88]. Also, a mixture of GAC, carbon cloth and nano-graphite was the optimal combination for better anaerobic of fat, oil, and grease waste [89]. In addition, conductive multi-walled carbon nanotubes (MWCNTs) were mixed with high-density polyethylene to formulate efficient conductive suspended carriers to be used in AD applications [90].

The addition of conductive additives to the anaerobic digesters has a lot of advantages to enhance the AD process where they can significantly increase the activity of methanogens through DIET [91,92]. They can also provide efficient metabolic pathways for high biodegradation of organic substrates during methanogenesis. In addition, they can provide large surface area for better biofilm formation and adsorption of organic materials [93,94]. They increase the richness and diversity of both archaeal and bacterial communities within the anaerobic digesters [95]. They also have relatively low cost and high mechanical strength properties [96], requiring less energy requirements [97].

Conductive functional materials allow high efficiency gains with respect to biogas production and solids destruction (Table 1). Multiple previous studies highlighted the significant influence of conductive functional materials on the enhancement of methane production [13,84,88], methane yield [39,85,98], VSS removal [84,99,100], and COD removal [90,100,101] during the AD of various feedstocks (Table 1). For example, Wu et al. 2022 showed that the cumulative methane increased by 12.28–62.91 % and the microbial activities within the digester were improved by 2.28–16.8 % when MWCNTs were used in AD of sewage sludge [90]. Moreover, wrapped GAC enhanced the TSS removal by 5.3 % in the digestion process of swine waste [84] while the powdered activated carbon resulted in an increase in the TSS removal by 36.4 % during the AD of sewage sludge [99].

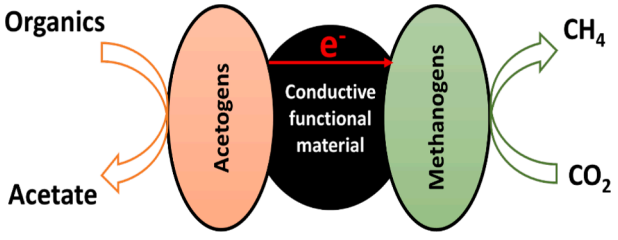


Fig. 4. Description of DIET between acetogens and methanogens using conductive functional materials in the anerobic digestion process.

There are some challenges associated with the addition of conductive materials in AD applications. For example, they might have a toxic effect on the microbes at high additive concentrations where they can adsorb the excess organic materials, causing inhibition of methane production by methanogens [113]. In addition, they are not ideal technology for treating complex wastes and substrates. In addition, although limited studies have examined the applicability of using conductive materials in pilot-scale systems [105], there are no previous investigation nor implantations of this technology in full-scale systems, hindering the feasibility of addressing this technology in full-scale applications. For big-sized carbon materials such as carbon cloth, they have lower specific surface area compared to particle-sized materials (e.g., GAC and carbon nano-tubes), leading to poor mass transfer and clogging of bed [90,114]. On the other hand, particle-sized materials can be easily lost in the discharge, resulting in production of higher sludge volume and complexity in operation and recovery [90].

2.3. Micro-aeration

Micro-aeration is an innovative, environmentally friendly and cost-efficient technology for enhanced production of methane in the AD process [115,116]. Micro-aeration depends on providing the anaerobic digesters with a limited amount of oxygen or air (Fig. 6). It remarkably improves the activities of acidogenic and hydrolytic microbes, resulting in improved hydrolysis of particulate substrates and slowly biodegradable components [117]. It stimulates the growth and activity of facultative microbes, excreting high amounts of extracellular enzymes to enhance the hydrolysis and fermentation which can improve the overall efficiency of the AD process [118,119]. It also leads to a higher accumulation of VFA and the conversion of short-chain fatty acids into acetate [120]. Moreover, the micro-aeration strategy plays a substantial role in removing hydrogen sulfide from biogas by inhibiting sulfate reducing bacteria [121,122]. Micro-aeration can be applied by either injecting the anaerobic bioreactor/digester with a limited amount of oxygen or air (dissolved oxygen concentration $\leq$ 1.0 mg/L) (Fig. 6) or applying the micro-aeration process as a pretreatment approach before the AD step in a separate reactor [121,123].

There are multiple factors that can affect the efficiency of micro-aeration technology. The oxygen dosage has a significant influence on the methane yield. While moderate micro-aeration can shorten the lag-phase, enhancing the methane yield, excessive and/or inappropriate micro-aeration can significantly inhibit the activity of methanogens, critically decreasing the methane yield [124,125]. Also, the micro-aeration rate and intensity can affect the solubilization of waste, hydrolysis reaction (mainly hydrolysis of proteins and carbohydrates) and methane yield [117]. Moreover, the organic loading rates, aeration contact time, loading, and frequency play an important role in the success of micro-aeration technology [126–128].

The efficiency gains from employing the micro-aeration technology in AD applications vary according to the wastewater sources (Fig. 7). In previous micro-aeration studies for AD of corn straw, it was emphasized that micro-aeration can increase the maximum methane yield by 10 % compared to the control reactors at different organic loading rates with substantial enhancement in the diversity and abundance of the bacterial and archaeal communities [125,126]. Moreover, the micro-aeration technique reflected better conductive environment and higher electron-transfer capacity because of the increased soluble humic acid concentration which enhanced the DIET pathways for the syntrophic methanogenesis [125,126]. Xu et al. (2014) demonstrated that the hydrolysis rate was enhanced by approximately 27 % for carbohydrates and 64 % for proteins using micro-aeration intensity of 258 L-air/kg-TS/day during the AD of food waste, while the COD removal and methane production were improved by 5 and 18 %, respectively [117]. Treating a mixed sewage sludge, micro-aeration increased the methane yield by

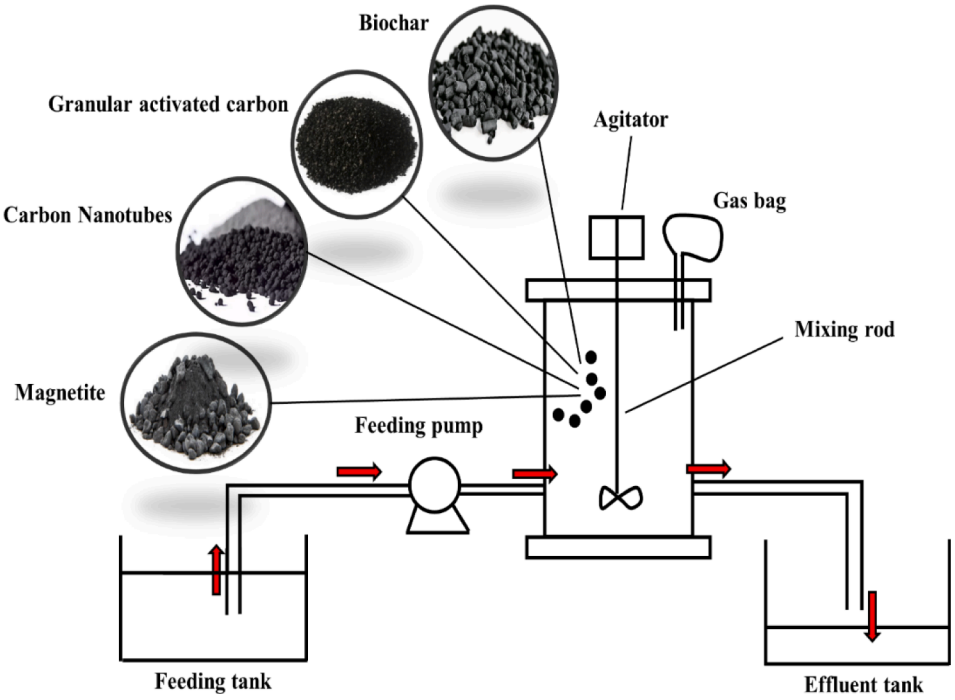


Fig. 5. Schematic diagram for the addition of conductive functional materials to anaerobic digesters.

Table 1
Comparison between efficiency gains from previous studies according to the carbon material and wastewater type.

Carbon material	Wastewater type	Methane production (%)	Methane yield (%)	Removal of VSS (%)	Removal of COD (%)	Reference
GAC	Synthetic Wastewater	31	–	–	–	[9]
GAC	Swine manure	10.6	–	6.6	–	[84]
GAC	Food waste	12.4	–	10.2	48.95	[102]
GAC	Sugar beet pulp	13.7	–	–	–	[103]
GAC	Lipid-rich wastewater	20 – 50	–	–	–	[13]
GAC	Synthetic Wastewater	16.7	–	–	–	[96]
GAC	Pharmaceutical Wastewater	11	–	–	13.4	[104]
GAC	Agro-industrial WW	32	–	–	–	[105]
GAC	WAS	50	–	–	25	[106]
GAC	Corn straw	–	34.37 – 38.15	–	–	[39]
GAC	Blackwater	–	38.3	–	–	[85]
GAC	Synthetic Wastewater	–	–	–	22.95	[107]
Biochar	Sewage sludge	16 – 25	–	–	–	[87]
Biochar	Mixed Sludge	20	–	–	–	[108]
Biochar	Sewage sludge	16.6	7.6	–	–	[109]
Biochar	Food waste	–	20.3	–	–	[110]
Biochar	Cow manure	–	5	–	–	[11]
Biochar	Swine manure	–	25	–	–	[12]
Magnetite	Food waste	–	30.1	–	–	[110]
Graphene	WAS	19.5	25	–	29.1	[101]
PAC	Sewage sludge	–	49	34.1	–	[99]
MWCNTs	Sewage sludge	12.28 – 62.91	–	–	3.6 – 37.2	[90]
Activated Carbon	WAS	30	–	–	–	[58]
Activated Carbon	WAS	14.3 – 34.1	–	–	–	[111]
Activated Carbon	Lignin waste	–	33	–	–	[10]
GAC & magnetite	FOG and Food waste	50 – 80	–	–	–	[88]
GAC and GBC	Wheat Husk and Sewage Sludge	–	22 – 27	–	–	[98]
Carbon Cloth / GAC	Dog food	–	–	22 – 50	45	[100]
Carbon Cloth / GAC	WAS	15.4	–	–	–	[74]
AC and Biochar	Piggery waste	6.9	–	–	3	[112]
Mix of GAC, carbon cloth, and nano-graphite	Fat, oil and grease waste	13–22	–	25 – 55	28 – 32	[89]

114 %, augmented soluble COD by 150 %, solubilization of proteins by 185 %, and solubilization of total sugar concentrations by 192 %, and reduced the VSS content by 40 % [129,130]. de Oliveira et al. (2021) observed that 0.17 L-O_2/L-feed/day was the optimal air dosage for a successful digestion of swine wastewater, yielding an approximate removal of VSS by 85 %, total COD by 84 % and particulate COD by 82 % [124]. Zeb et al. (2022) reported that micro-aeration with contact time of 12 h and single injection of 7.3 mL-O_2/g-VS for buffalo manure

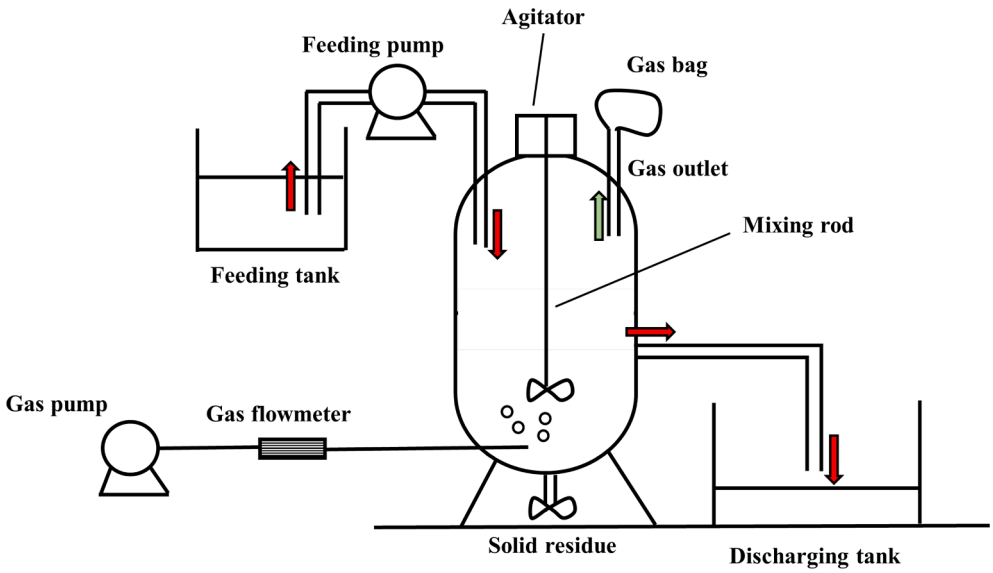


Fig. 6. Schematic diagram for the micro-aeration technology in anaerobic digesters.

digestion enhanced the acid-soluble lignin degradation by 78 %, resulting in an increase of 32 % in the average daily biogas yield [127]. Micro-aeration process of corn straw coupled with thermophilic conditions increased the maximum methane yield by approximately 17 %, the reduced sugar by 9 %, the soluble COD by 5 % and volatile fatty acids by 24 %, enhancing the disintegration of particulate and slowly biodegradable compounds [131,132]. Valero et al. (2020) reported that micro-aeration increased the production of VFAs by 62 % which promoted the DIET process, leading to an increase in the methane yield by 55 % [133]. In other previous studies, the removal efficiency of the volatile solids in AD ranged from 18 to 54 % using the micro-aeration technology while both specific biogas production and methane yield reached to approximately 17 % [116,122,123,134].

One of the main challenges of implementing micro-aeration in full-scale systems is the automatic process control of the micro-aeration process where the management of automatic online data acquisition schemes and accurate supervisory control systems for appropriate dosing of oxygen are essential for successful operation using micro-aeration. In addition, there are significant difficulties in the online measuring and testing of multiple physicochemical parameters (e.g., microbial growth and activity), hindering the implementation of micro-aeration technology in full-scale intensification applications [2].

2.4. *Anaerobic membrane bioreactors (AnMBRs)*

Micro- and ultra-filtration membranes can be coupled with anaerobic digesters for efficient AD processes where membranes can retain the biomass in the digester for higher biomass concentrations. Whereas the filtered liquid (permeate) is collected after it passes through the membrane which results in effective decoupling of HRT and SRT [135]. There are three different configurations of the anaerobic membranes (AnMBRs): submerged/immersed, externally submerged and side-stream (crossflow) configurations (Fig. 8). For submerged AnMBRs, the membrane modules are directly submerged in the sludge inside the bioreactor, requiring less footprint and energy requirements where the permeate is driven by gravity or a pump (Fig. 8a). For externally submerged AnMBRs, the membrane modules are submerged in an external tank, where sludge is circulated between the digester and the external tank, which provides easier maintenance without a need to open the anaerobic bioreactor (Fig. 8b). For the side-stream AnMBRs, the membrane modules are placed outside the bioreactor where the sludge is circulated at relatively high velocity through a recirculation loop (Fig. 8c). Although the crossflow over the membranes surface allows sufficient physical cleaning for the membranes surface and better fouling control, the energy requirements for effective cross-flow velocity are high [136].

There are various types of the membranes that can be used in AD and intensification applications (Table 2). These types can be divided into conventional and modified membranes. Each type of these membranes has different characteristics, advantages and disadvantages which makes the selection of membrane type highly dependent on the type of sludge. For example, expanded granular sludge bed reactor (EGSB)-AnMBR is preferable for treating domestic wastewater and wastewater that contain elevated concentrations of lipids and toxic compounds [137,138]. While anaerobic fluidized-bed membrane bioreactor (AFBR)-AnMBR is highly recommended for the treatment applications related to pharmaceuticals [139]. The AnMBRs can be combined with anaerobic ammonia oxidation (Anammox) process in Anammox-AnMBR system to allow simultaneous nitrogen removal and methane production [140,141]. In addition, AnMBRs can be integrated with forward osmosis system, formulating AnOMBR system, for efficient organics degradation, complete removal of total phosphorus and partial removal of ammonia and total nitrogen [3,142]. Moreover, anaerobic membrane sponge bioreactor (AnMSBR)) is considered as optimal choices for limited membrane fouling systems [137,138,143,144] while anaerobic dynamic membrane bioreactor (AnDMBR) and anaerobic osmotic membrane bioreactor (AnOMBR) allow high accumulation of VFA, ammonia, and salts [137,145–147].

It should be noted that multiple inhibitors can negatively affect the production of biogas by AnMBRs such as ammonia, long chain fatty acids (LCFAs), salts, and sulfide. Elevated concentrations of ammonia can decrease the methane yield below the theoretical value and deteriorate the activity of methanogenic microbes. Ammonia inhibition also reflects higher energy requirements, instability in pH conditions, inhibition in enzymatic activity and critical damage to the cell membrane of microorganisms [160]. Although LCFAs are primary substrates in biogas production, their toxicity can limit substrate mass transfer, decrease the methanogenic activity, and impair sludge washout and flotation especially during high-strength-lipid treatment process [161,162]. Salts are considered common substrates where high salt concentrations are highly associated with food, petroleum, chemical and pharmaceutical wastewater. High salinity plays a crucial role in the dehydration of the bacterial cells because of the osmotic pressure, negatively influencing the methane production. In addition, elevated salinity levels can cause

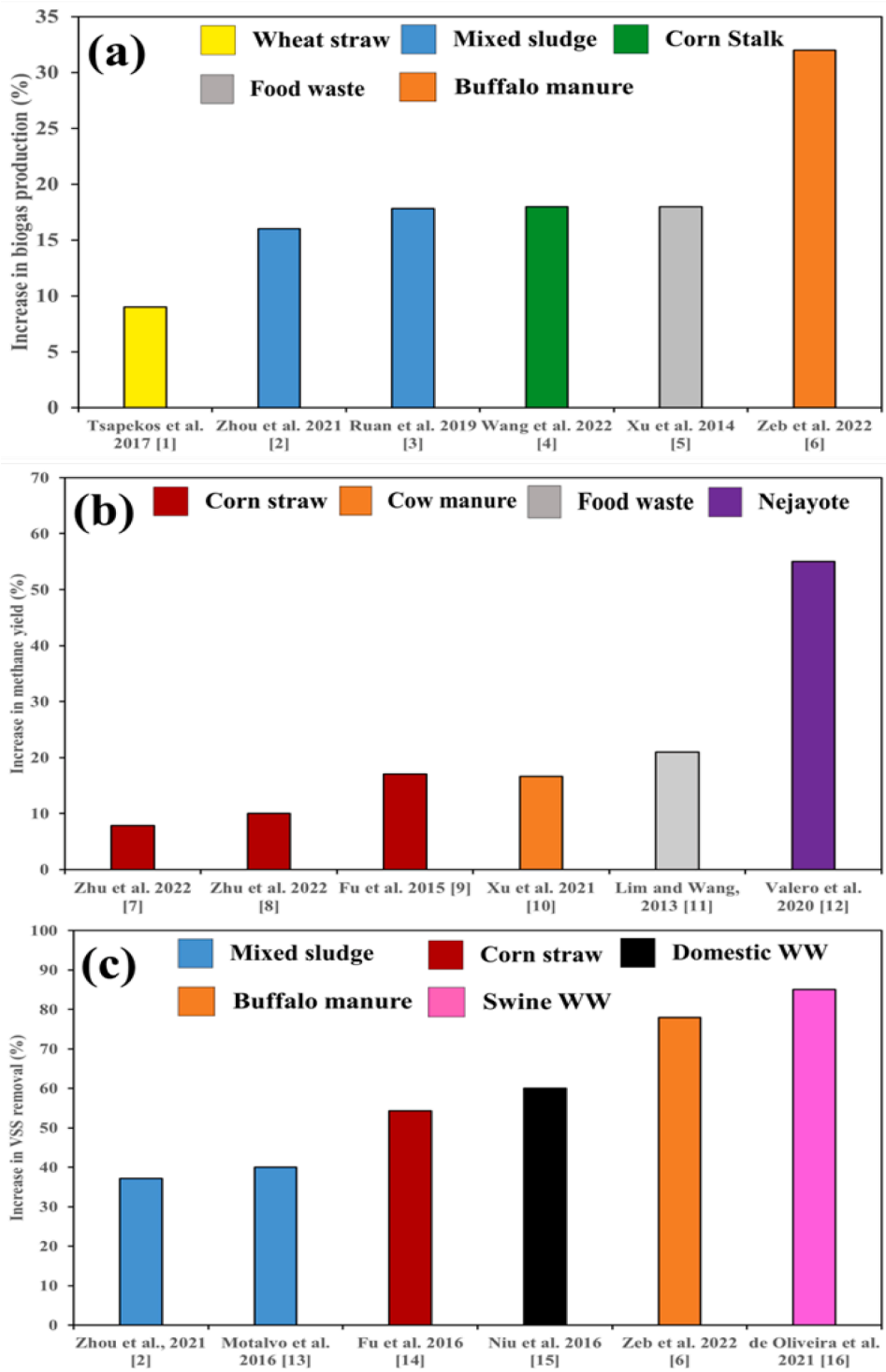


Fig. 7. Efficiency gains due to the application of micro-aeration technology in previous AD studies through increasing: (a) biogas production, (b) methane yield, and (c) VSS removal. (All references used for preparing the figure are listed in the supplementary material section.).

microbial inactivity and enzyme inhibition, decreasing the methane yield [144].

Generally, there are multiple advantages linked to the integration of membranes with AD processes (Table 2). For example, AnMBRs can sustain high cell concentrations at relatively high hydraulic loading and sufficient mixing conditions because of the complete decoupling of HRT and SRT [163]. In addition, they have relatively high toxic resistance and higher biogas productivity [164]. In addition, AnMBRs allow simultaneous organics and nutrient removal with high biogas production rate [165]. AnMBRs allow ambient-temperature AD process through increasing the SRT, reducing the total cost of the system and the negative influence on the environment and society [166]. In addition, long SRT in membranes allows better stabilization of the produced biosolids, reducing the sludge production rate and its associated disposal cost [136]. Ultimately, AnMBRs significantly reduces the greenhouse gas emissions since it requires less energy consumption with high-quality methane that can be recovered and used as an energy source.

The total life cycle capital cost of AnMBRs may include membranes,

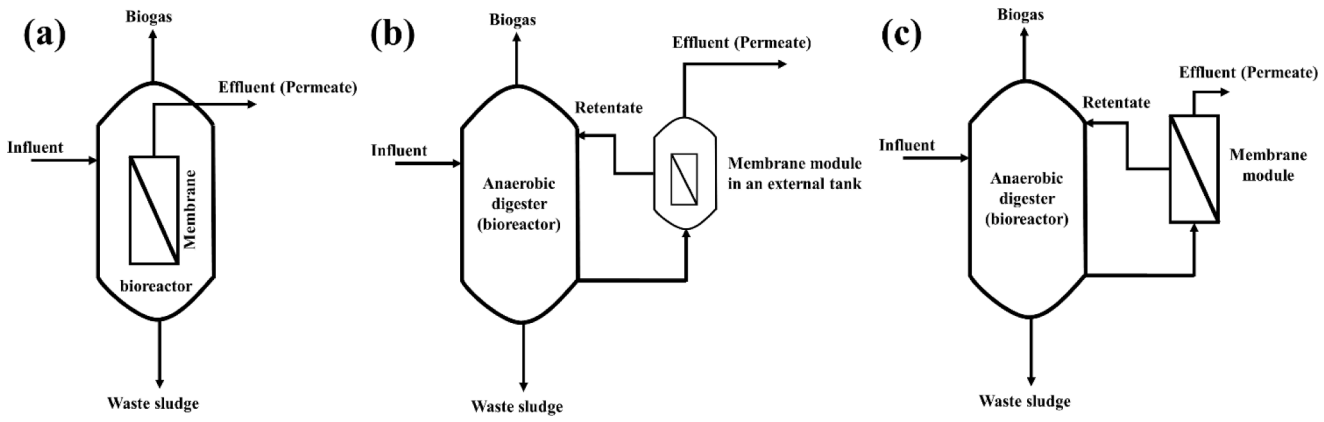


Fig. 8. Schematic diagram for the configurations of AnMBRs: (a) immersed/submerged, (b) externally immersed/submerged, and (c) side stream configurations.

tank installation, screens, and gas blower costs [167]. The major cost is associated with the membranes (≅ 73 %) and the tank installation cost can contribute with approximately 12 % of the total capital cost while the rest of the cost is divided between screens, gas blowers and other costs [168]. Regarding the operational cost, approximately 47 % of it is invested in the gas scouring energy while 33 % of this cost can be allocated for chemical consumption. The rest of the operational cost can be divided into 13 % for pumping energy and 7 % for sludge disposal [168]. The previously mentioned fractions of the capital and operational costs are determined for a submerged anaerobic membrane system that treats approximately 20,000 m^3/day of municipal wastewater [168]. In a previous study that investigated the performance of an AnMBR in the treatment of sulfate-rich urban wastewater, it was emphasized that the energy consumption of biogas recycling blower is the highest among other energy consumers where it represented approximately 74 % of the total energy consumption [169]. While the sludge feeding pump, stirring power reactor and permeate pump consume approximately 15, 8 and 3 % of the total available energy, respectively.

There are multiple challenges associated with the full-scale AnMBRs (Table 2). The performance of AnMBRs declines under harsh conditions such as thermophilic temperature, high level of toxicity, high organic loading rate and high salinity, negatively affecting the biogas production rate [137]. Furthermore, membranes require high installation, capital and operating costs, hindering the adoption of this technology in full-scale AD applications [167]. Membrane fouling is also a crucial challenge that is associated with the application of AnMBRs [170], decreasing the membrane permeability and further the system productivity.

2.5. Application of hydrogen injection

The hydrogen injection process is responsible for converting the carbon dioxide (CO_2) in anaerobic digesters into methane through a hydrogenotrophic methanogenic reaction that is called the "Bio-Sabatier Process" ($4H_2 + CO_2 \rightarrow CH_4 + 2H_2O$) (Fig. 9). There are two possible configurations for the injection of hydrogen to anaerobic digesters, in-situ and ex-situ configurations. For the in-situ configurations, the hydrogen is injected directly into the digesters [171]. While for the ex-situ configurations, hydrogen and biogas from anaerobic digester are mixed and fed to an external anaerobic bioreactor for biogas upgrading [172].

Several factors control the performance of the AD process when hydrogen is injected into the digesters. The hydrogen mass transfer from the gas phase into the liquid one is considered one of the most crucial parameters for efficient hydrogen addition into anaerobic digesters [173]. It is highly dependent on the volumetric mass transfer coefficient and hydrogen concentration gradient between the two phases [173]. In addition, hydrogen partial pressure is a critical parameter that affects the activity of methanogens and fermentation bacteria [174]. Finally, the hydrogen-to-carbon dioxide ratio (H_2/CO_2) plays a significant role in enhancing the performance of anaerobic digesters using hydrogen injection techniques [171,175].

One of the main advantages of in-situ hydrogen injection is that the hydrogen is injected/added directly into the process liquid, allowing better utilization of hydrogen and biogas production by hydrogenotrophic methanogens [171,176,177]. In addition, the direct injection process excludes the necessity of having additional reactors since the methane enrichment process occur within the same reactor, reducing the capital cost and footprint requirements of the treatment facility [178]. As a result, the in-situ hydrogen addition technology can be integrated into existing facilities and infrastructure in a direct and easy way [175]. On the other hand, the ex-situ hydrogen injection technology involves the usage of another reactor to be fed with hydrogen and biogas, increasing the capital cost of implementing this technology in full-scale treatment facilities [172]. Hydrogen addition techniques can also increase the calorific biogas since they elevate the methane content in the biogas in anaerobic digesters [179–181]. This would lead to a significant increase in the energetic content of the produced biogas [174].

Using the hydrogen addition technology, the methane content in biogas can increase from approximately 55 – 60 % (i.e., average methane concentration from typical AD) to more than 95 % [182,183]. In a previous pilot-scale study, it was demonstrated that hydrogen injection enhanced the concentration of gas phase methane by 36 % compared to the theoretical maximum concentration from the glucose anaerobic degradation during continuous tests where the methane content was initially increased by 68 % due to the activity of hydrogenotrophic methanogenesis [174]. Kim et al. (2021) emphasized that the methane content in biogas was increased to higher than 90 % when the hydrogen injection was integrated with auto-generative high-pressure digestion of food waste [180]. Zhu et al. (2020) observed that the maximum average methane yield increased by approximately 30 % when hydrogen was injected into anaerobic digester of swine manure waste [179]. In another previous study, the range of average methane content in the biogas was 65 – 72 % when hydrogen was added to enhance the AD of cattle manure [178]. It was also noted in another study that the cumulative methane yield increased by 56 % when hydrogen was added to a municipal anaerobic digester with an H_2/CO_2 of 4:1 while the methane content in the biogas reached approximately 90 % [171]. Furthermore, Martínez et al. (2019) stated that hydrogen addition increased the biogas production in sewage sludge by 12 % compared to the control reactors [184].

There are some challenges associated with the application of hydrogen injection approach in the AD process. Hydrogen injection is coupled with the increase of pH inside digesters due to CO_2 depletion, in which the pH may increase to more than 8.5 when the hydrogen

Table 2
Comparison between different types of anaerobic membranes according to the characteristics, advantages, and disadvantages.

Membrane	Abbreviation	Category	Characteristics	Advantages	Disadvantages	References
Completely stirred tank reactor	CSTR-AnMBR	Conventional	• Most common AnMBR. • Cylindrical or rectangular tank. • Mechanical mixing conditions.	• Less mass transfer resistance. • High substrate-biomass contact. • High biomethane potential	• Rapid acidification. • High VFA inhibition. • Low biomethane potential associated with low organic loading rates. • Less activity of specific methanogenic activity.	[148,149]
Upflow anaerobic sludge blanket	UASB-AnMBR	Conventional	• Granular-based reactor. • Cylindrical or rectangular column. • Sufficient mixing conditions.	• Reasonable substrate-biomass contact. • High hydraulic/organic loading rates. • Less gas sparging demand. • No mechanical mixing. • Moderate toxic resistance.	• Long start-up period. • Poor biogas production and system instability when hydraulic selection pressure is eliminated. • Poor mixing at psychrophilic conditions.	[138,150,151]
Expanded granular sludge bed reactor	EGSB-AnMBR	Conventional	• Granular-based reactor. • Tall column reactors. • Small footprint. • High Upflow velocity. • High mixing conditions.	• Reasonable substrate-biomass contact. • Enhanced mass transfer. • Better than UASB in mixing and temperature limits. • Effective in domestic and lipids and toxic compounds-rich wastewater.	• High possibility of sludge washout and granules fragmentation. • Granulation is not common.	[137,138]
Anaerobic fluidized-bed membrane bioreactor	AFBR-AnMBR	Conventional	• Granular activated carbon-based reactor. • Tall column reactors. • Small footprint. • Two-stage reactor. • High surface area per unit volume of reactor.	• High inhibitors and toxic resistance. • High hydraulic/organic loading rate. • Low capital cost and reactor size. • Effective in domestic wastewater. • Effective in treating pharmaceuticals.	• Two-step process. • Existence of dissolved methane. • VFA accumulation at short HRT. • Low activity of Acetoclastic methanogenesis.	[139,146,152]
Jet flow anaerobic bioreactor	JFAB-AnMBR	Conventional	• Jet flow system. • Inner tube and nozzle system for mixing.	• Sufficient mixing. • Appropriate reactor size and design. • Sound homogenization inside the reactor.	• Poor in dealing with toxic compounds. • VFA accumulation.	[137]
Anaerobic ammonia oxidation	Anammox-AnMBR	Modified	• Combination between Anammox reactor and AnMBR. • Simultaneous nitrogen removal and methane production. • Homogeneous distribution of biomass and substrates.	• Cost and energy efficient. • Less membrane fouling. • Small footprint. • High nitrogen removal. • Short start-up time.	• Slow Anammox process. • Existence of dissolved methane. • Accumulation of nitrogen gas within biogas. • Ammonia inhibition for methane production. • Existence of dissolved methane.	[140,141]
Anaerobic dynamic membrane bioreactor	AnDMBR	Modified	• Solid-liquid separation process. • Low cost supporting materials (e.g., meshes and macropores fabrics).	• High membrane flux. • High biogas production. • Low cost of membrane modules. • Less membrane fouling. • Less energy requirements.	• Biogas sparging. • High shear stress. • Methane oversaturation in the permeate. • Possible ammonia inhibition. • Potential biogas escape from the external membrane unit.	[138,145,153,154]
Anaerobic membrane distillation bioreactor	AnMDBR	Modified	• Thermal-driven separation process. • High SRT compared to HRT. • Mixing by gas sparging.	• Small footprint. • High biogas production rate. • Complete removal of total phosphorus. • Less methane loss in liquid phase. • Short start-up time.	• Mandatory post-treatment to recover ammonia and methane. • High heat requirements.	[155,156]
Anaerobic osmotic membrane bioreactor	AnOMBR	Modified	• Combination between forward osmosis and AnMBR. • Draw solution requirements. • High injection capacity.	• High contaminant degradation. • High biogas production rate. • Reasonable energy recovery. • High total phosphorus removal. • Moderate ammonia and total nitrogen removal.	• High accumulation of salts and ammonia. • Possible pH instability. • Disposal of inorganic-rich supernatants. • Low tolerance of membranes to high temperatures.	[142,147]

(*continued on next page*)

Table 2 (*continued*)

Membrane	Abbreviation	Category	Characteristics	Advantages	Disadvantages	References
Anaerobic membrane sponge bioreactor	AnMSBR	Modified	• Polyurethane sponges as bacteria carrier. • Single or two-stage process. • Rotary disk-supporting media.	• Cost efficient option. • High organic and nutrient removal. • Less membrane fouling.	• Lack of knowledge about the effect of sponge size, shape, and material. • Limited data and applications in literature.	[137,157]
Gas-lifting anaerobic membrane	Gl-AnMBR	Modified	• Combination of low-pressure membrane filtration and AnMBR. • Two-stage process.	• Minimal energy requirements. • Minimal dissolved methane in the permeate. • Cost efficient system.	• High shear force. • Low biogas production. • Less activity of specific methanogenic activity. • Possible methane oversaturation and pH changes. • Mandatory helium gas for start-up.	[137,158]
Vibrating anaerobic membrane	V-AnMBR	Modified	• Vibration instead of gas-sparging for membrane fouling control. • No mixing.	• Less membrane fouling. • No biogas sparging.	• Unknown effect of shear stress by vibration on the microorganisms. • VFA accumulation. • Decline is methane production.	[143,159]
Anaerobic bio-entrapped membrane bioreactor	AnBEMR	Modified	• Trapping of biomass in the bio-carriers/ball.	• Suitable for high biomass concentrations. • Less membrane fouling. • Simultaneous removal of carbon and nitrogen. • Effective on complex organics. • High organic loading.	• Accumulation of toxic organics. • Poor at high salinity. • Longer filtration and start-up time. • Limited data and applications in literature.	[137,144]

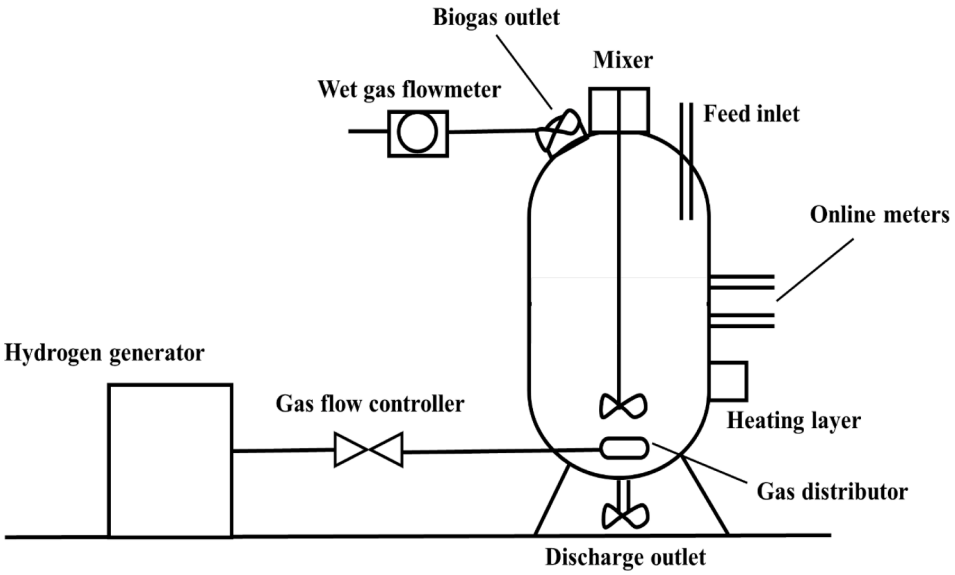


Fig. 9. Schematic diagram for the hydrogen injection/addition technology in anaerobic digesters.

injection rate is increased. Moreover, at high hydrogen injection rates, the residual hydrogen can hinder the AD process since the syntrophic reactions can be inhibited when hydrogen exists in the anaerobic digesters [180]. Hydrogen injection can also increase the hydrogen partial pressure, inhibiting the syntrophic acetogens leading to accumulation of VFAs [181]. In addition, the high cost of hydrogen is one of the most critical drawbacks of applying hydrogen injection technology in the AD process [185].

2.6. IntensiCarb (Vacuum-assisted AD)

IntensiCarb^{TM} (IC) is a patent technology that was jointly developed by USP Technologies, Brown and Caldwell and Western University. IC is an innovative technology for vacuum-enhanced biochemical fermentation and AD using mild vacuum pressure ($\cong$ 110 mbar) at temperature of 20 – 60 °C using a highly efficient solid–liquid separation process [186]. The IC is considered a single treatment step since it combines multiple AD processes within the same compartment [187]. In other words, thickening, hydrolysis, acidification, dewatering, and gas stripping are integrated into a single bioreactor in the IC technology. In addition, the IC admits complete decoupling of HRT from SRT through vacuum evaporation [188]. It is also considered as a compact process that can be implemented on different facility scales including relatively small municipalities ($\leq$ 10 mgd). There are multiple advantages for the IC technology since it allows simultaneous enhancing sludge thickening and assists in resource recovery and separation such as phosphorus and ammonia [186]. Moreover, at acidic conditions (i.e., pH $<$ 6), nitrogen can be retained in the fermentate while it can be stripped and transferred to the condensate at alkaline conditions (i.e., pH $\geq$ 8).

The IC reactor consists of a heat exchanger to allow the pre-heating

of the feedstock by the heat recovered from the fermented sludge (Fig. 10). It has also another heat exchanger to recover and recycle the latent heat of evaporation within the system. In addition, the IC has a main reactor vessel that can be operated under vacuum to accomplish the thickening and fermentation processes. It contains a vacuum pump to extract the produced vapor during the evaporation process. Additional information and details about the IC technology can be found in Haroun et al. (2022) and Okoye et al. (2022) [186,188].

Haroun et al. (2022) compared between the IC technology and conventional fermenter to evaluate the feasibility of implementing the IC in the fermentation applications and assess the effect of vacuum on the activities of the involved microorganisms [186]. It was demonstrated that the IC achieved an increase in VFA generation with a range of 29.5 – 90.2 % while the hydrolysis yield was increased by 16.2 – 56.4 % compared to the conventional fermenter. It was also emphasized that the vacuum did not have a negative influence on the activity of acidogenic bacteria and fermenters [186]. Okoye et al. (2022) found that the IC could enhance particulate solubilization by 65 % and allow total ammonia extraction by approximately 50 % [188]. Khadir et al. (2024) used an IC system with an external vacuum evaporation unit for sludge digestion and compared its performance with a conventional digester [189]. It was reported that using IC would allow the application of organic loading rates that are three times higher compared to conventional digesters without any deterioration in the methane yield. In addition, approximately 49 % of nitrogen in the influent sludge was recovered in the form of ammonia in the condensate using the IC system. These experimental outcomes confirm the validity of using the IC as an intensification technique in the AD and fermentation applications in the lab-scale systems.

2.7. Microbial hydrolysis process (MHP) using C. Bescii.

In the AD processes, hydrolysis is considered as the rate-limiting process on efficient organics solubilization and biomass destruction [26,190,191]. Therefore, some research investigations were oriented towards enhancing the hydrolysis reaction using the microbial hydrolysis process (MHP) that can convert the organic wastes into biogas through a two-step process [192]. The first step involves the pre-digestion process using hyper-thermophilic anaerobic bacteria, such as *Caldicellulosiruptor bescii, Caldicellulosiruptor saccharolyticus*, and *Thermotoga maritima* (Fig. 11). The second step is a conventional AD process which contains a mixed culture from the effluent of pre-digestion step and microorganisms responsible for acidogenesis, acetogenesis and methanogenesis [192]. It should be mentioned that the feedstock from MHP needs different methanogens' consortium compared to the conventional AD because the microorganisms in the AD feed (i.e., pre-digest effluent) are unsuitable for continuous seeding of the AD system with the required bacteria and methanogens. In addition, MHP pasteurizes the feedstock, leading to more depletion in the methanogens and other associated microbes under continuous seeding conditions for the AD system. Therefore, the anaerobic digesters must be seeded at the beginning with methanogens that match the microbiology of the pre-digestion effluent. Moreover, the retention times should be sufficient to allow better growth and activity of the consortium. As a result, the MHP can achieve up to 85 % destruction of the biomass (mainly volatile solids) while the percentage of methane in the produced biogas can reach to 75 % [193].

The MHP is considered an innovative and effective technology that can deal with multiple recalcitrant wastes (e.g., manure, grass, dairy and woody wastes). These wastes have relatively large organic solids and contain polymeric molecules (e.g., lignocellulosic materials) that are not easily accessible to methanogens [28,43]. In the MHP, *C.bescii* can change the raw characteristics of these wastes into a more readily digested mixture of small, water-soluble ions, sugars, and molecules, enhancing the efficiency and rate of substrate conversion into biogas in the AD process [192]. *C.bescii* is a hyper-thermophilic obligate anaerobic bacterium that can grow at neutral pH conditions (i.e., pH=7 – 8) and temperature of 70 to 80 °C [194,195]. *C.bescii* is responsible for the effective degradation and solubilization of recalcitrant wastes such as lignocellulosic materials in dairy waste and switchgrass in plant biomass since *C.bescii* can produce more than 50 enzymes for efficient hydrolysis reactions [192,193,196,197]. *C. bescii* can formulate a group of hydrolase enzymes to hydrolyze proteins and esters into water-soluble small molecules and ions. They can also depolymerize lignin and hydrolyze multiple polysaccharides (e.g., hemicellulose, xylose, and crystalline cellulose) into water-soluble C5 and C6 sugars that can be catabolized by *C.bescii* into lactate and acetate [195,198]. These produced substrates can be easily consumed by the Acetoclastic methanogens for active methanogenesis process and conversion into biogas.

Hansen et al. (2021) demonstrated that the *C. bescii* was an effective hyperthermal anaerobic bacterium for enhanced waste solubilization during the pre-digestion step for multiple types of wastes including dairy and manure wastes on lab- and pilot- scale systems [193]. It was also found that the pre-digestion using *C. bescii* increased the conversion rate of waste into biogas by two times compared to the control scenario (with no pre-digestion) in the lab-scale tests, confirming the efficiency of the

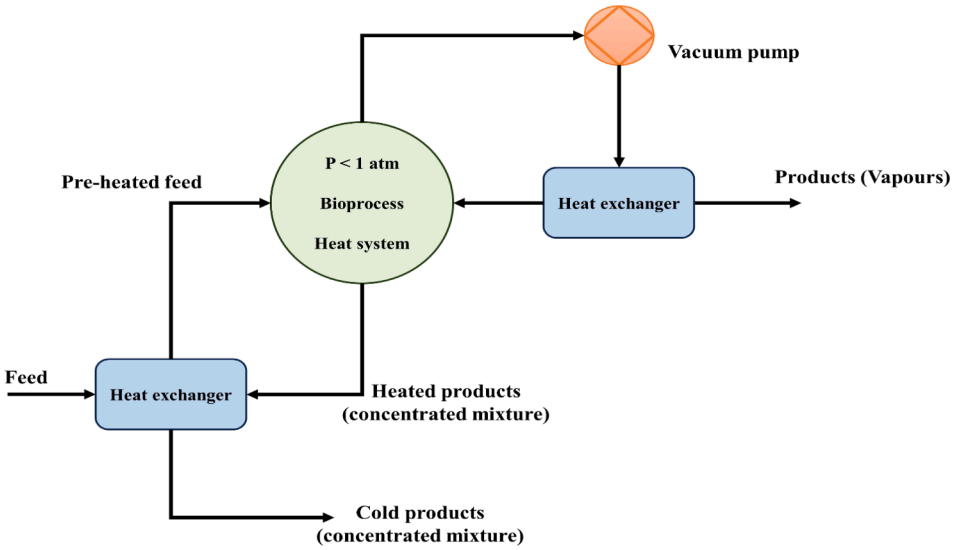


Fig. 10. Schematic flow diagram for the IntensiCarb technology in the AD process.

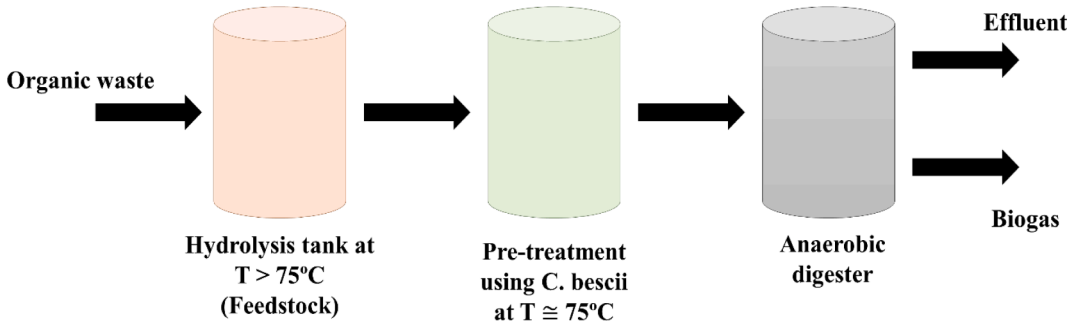


Fig. 11. Schematic diagram for the main three steps in MHP using *C.bescii* technology in the AD process.

pre-digestion process using *C. bescii*. In a pilot-scale system, the destruction of volatile solids reached up to 85 % while the methane content in the biogas was approximately 75 %, showing a high potential of using *C. bescii* in engineering commercial-scale systems [193].

C. bescii can also be used in dark fermentation applications where it can be co-cultured in many cases with other extreme thermophiles such as *Caldicellulosiruptor saccharolyticus* and *Thermotoga maritima*. The co-culture of *C. bescii* and *C. saccharolyticus* could enhance the degradation of garden waste and increase the hydrogen production yield compared to the individual cultures [199]. This process resulted in producing biohythane (hydrogen-enriched biogas) that contains approximately 15 % of hydrogen from garden waste, yielding a high potential of energy generation and recovery. Other studies observed an increase in the hydrogen production rate when *C. bescii* was mixed with wastewater biosolids in the dark fermentation process of cattle manure [198,200].

C. bescii is considered an efficient candidate for bioaugmentation in a combined steam-explosion (SE) pretreatment and bioaugmentation using cellulolytic bacterium because *C. bescii* can utilize numerous substrates, such as hemicellulose, cellulose and lignocellulosic biomass and ferment C5 and C6 sugars simultaneously [195]. *C. bescii* can also utilize multiple cellulolytic enzymes with different binding and catalytic domains during the bioaugmentation process unlike various other cellulolytic bacteria [201]. In a previous lab-scale study, it was emphasized that the methane yield was increased by 140 % using the SE and bioaugmentation processes [190]. The maximum methane production rate was also increased from 4 up to 14 mL-CH_4/g-VS when the SE and bioaugmentation processes were applied compared to the untreated condition. In addition, the initial methane production rate at the beginning of processes was increased by 21 – 44 % compared to the condition without bioaugmentation, confirming the positive contribution of bioaugmentation using *C. bescii* on the biogas production rate.

There are multiple challenges that hinder the implementation of MHP using *C.bescii* technology in full-scale AD applications. This technology is associated with high capital cost for employing the three stages of treatment (i.e., hydrolysis, pretreatment, and AD) (Fig. 11). Also, high operational costs and energy requirements are required for heating and pretreatment processes. In addition, *C.bescii* are sensitive to the operational conditions such as temperature and pH. During the hydrolysis of complex substrates, multiple inhibitory compounds (i.e., phenols, fatty acids) and/or toxic intermediates could be released, negatively affecting the activity of *C. bescii* and other microorganisms involved in the AD process. In many cases, low hydrogen yield per mass of substrate is noticed in the MHP using *C.bescii*.

3. Comparison between different innovative intensification technologies

According to the review of the studies on innovative technologies for intensification in literature (Section 2), it can be concluded that some technologies are well-studied on the lab-scale investigations such as MEC-assisted AD, micro-aeration, AnMBRs, and addition of conductive materials. Some of these technologies, such as IntensiCarb and MHP by *C.bescii*, still require further investigations to determine their governing parameters and optimal operational conditions due to a lack of knowledge about various aspects of these processes.

According to the efficiency gains of the most common three innovative technologies (i.e., MEC-AD, micro-aeration, and conductive functional materials) from multiple studies in literature (used for preparing Fig. 12), it was emphasized that MEC-AD technology results in higher methane production and yield compared to micro-aeration and conductive functional materials (Fig. 12a and c). Methane production reflects the total amount of methane gas produced during the AD process which is usually expressed in terms of volume (e.g., liters) of methane generated [202]. It can also be presented in terms of the rate of methane produced per unit of time (e.g., L/day). On the other hand, methane yield refers to the amount of methane produced per unit of substrate (e.g., COD and TS) added to the AD process which is typically expressed as volume of methane produced per unit mass of substrate (e.g., L-CH_4/kg-COD) [202,203]. In general, methane production gives an indication of the overall methane generation capacity of a digester over a specified period while methane yield is used to measure the efficiency of the substrate conversion into methane during the AD process [202,203]. The approximate range of increase in methane production was 20 – 60 % for MEC-AD, 15 – 22 % for micro-aeration, and 14 – 32 % for the conductive functional materials (Fig. 12a). Similarly, the increase in methane yield ranged from approximately 10 – 60 % for MEC- AD, 10 – 30 % for micro-aeration, and 22 – 37 % for the usage of conductive functional materials (Fig. 12c). On the other hand, MEC-AD technology yielded in lowest enhancement in VSS and COD removal among the other three common innovative technologies with an approximate range of 10 – 38 % for the increase in VSS removal, and 10 – 20 % for the increase in COD removal (Fig. 12b and d). Although there were no recorded efficiency gains in the increase of COD removal for the micro-aeration technology, its increase in VSS removal was the highest among the three common technologies where the improvement in VSS removal varied from 40 – 78 % (Fig. 12b). Regarding the addition of conductive functional materials, the increase in VSS removal ranged from approximately 10 – 50 % while the approximate range of increase in COD removal was 15 – 38 % (Fig. 12b and d). Therefore, it is highly recommended to implement the MEC-assisted AD technology in the intensification applications that involve higher methane production and/or yield requirements. For better removal of VSS and COD, it is highly recommended to use the micro-aeration and/or addition of conductive functional materials for better solubilization and destruction of sludge.

Given the main insights from studies in the literature that investigated multiple innovative technologies for intensification, a comparison between the advantages and challenges associated with each technology was constructed for better guidance about the appropriate technology according to the proposed budget and constraints in resources (Table 3). The selection of the innovative intensification technology is highly dependent on the nature and complexity of sludge and substrates. For

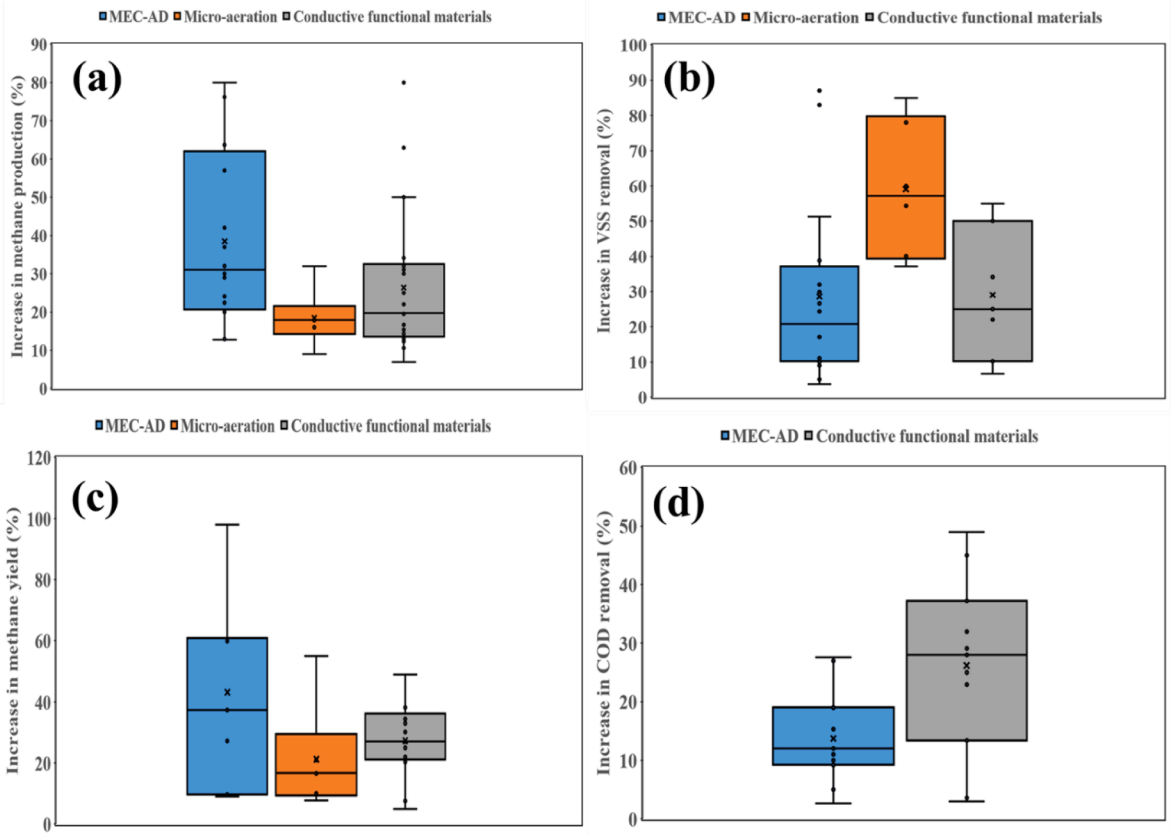


Fig. 12. Comparison between the efficiency gains of the three most common technologies in literature according to the increase in: (a) methane production, (b) VSS removal, (c) methane yield, and (d) COD removal.

example, MHP using *C.bescii* can be considered an effective technology for degrading the recalcitrant wastes (e.g., manure, wood, and dairy wastes) that contain elevated levels of organic solids and polymeric molecules [28,43]. This is mainly because *C.bescii* can change the raw characteristics of these substrates into a more readily digested mixture of small molecules and ions [192]. In addition, *C.bescii* can produce multiple thermostable enzymes for higher hydrolysis reactions [193,197]. Moreover, micro-aeration and MEC-AD technologies are suitable candidates for applications that require high solubilization of complex substrates and slowly biodegradable components since they can enhance the activities of acidogenic and hydrolytic microorganisms [17,78,116,117].

From an economic perspective, it can be observed that high capital and operational costs are highly linked with the implementation of multiple innovative technologies. For example, high capital costs (e.g., construction, electrodes, and membranes) and operational costs (e.g., electrical energy input) are required for the effective implementation and operation of MEC-AD technology in the intensification process [17,46,78]. Also, there is a high operation and maintenance cost associated with the membrane fouling in the AnMBRs [170,204]. In addition, the cost of hydrogen is relatively high when the hydrogen injection/addition technology is applied in intensification applications [185]. However, the hydrogen injection technology requires less capital costs and footprint size for the treatment facility [178]. According to the comparison between the different innovative intensification technologies, multiple cost-efficient technologies have low capital costs, such as micro-aeration technology [2] and low energy requirements such as conductive functional materials [96,97].

It should also be noted that the addition of conductive functional materials, hydrogen injection (in-situ configuration), AnMBRs (especially the immersed configuration), and micro-aeration technologies can be easily integrated with the existing infrastructure and current AD facilities (Fig. 5, Fig. 6, and Fig. 9), maximizing the feasibility of testing these technologies in full-scale intensification applications. Some technologies, such as MEC and membranes, can be immersed/submerged inside the bioreactor (i.e., internal configuration) for practical and simple integration with the current infrastructure and facility systems (Fig. 1, and Fig. 8). On the other hand, the implementation of MHP using *C.bescii* technology requires two additional steps (i.e., hydrolysis and pretreatment) before the sludge is being transported to the anaerobic digester (Fig. 11), increasing the capital and operational costs of this process.

4. Integration of intensification technologies with other technologies

In the literature, the MEC-AD systems were integrated with multiple innovative technologies for improving the methane productivity and microbial communities within the AD systems (Fig. 13). MEC-AD was combined with ultrasound, alkali, thermo-microaeration technologies to break the walls of bacterial cells and disintegrate the flocs of WAS, releasing high amounts of soluble organic matter [205,206]. Bao et al. (2020) reported that the methane production rate increased by three times compared to the control reactor while the methane yield was enhanced by approximately 98 % after pre-treating WAS by ultrasound-alkali [205]. In addition, the efficiency of VSS removal was approximately 25 % while the COD removal efficiency reached to 15.30 % [205]. MEC-AD technology can also be combined with thermophilic pretreatment to enhance the solubilization of organic compounds. In previous studies, it was demonstrated that the efficiency of COD removal can reach up to 29 % while the VSS removal efficiency ranged from 83.24 to 87.46 % (according to the applied voltage and organic loading rate) when the MEC-AD system was integrated with thermophilic pretreatment [61,207]. Chang and Champagne (2022) emphasized that the

Table 3
Comparison between advantages and disadvantages of the innovative technologies.

Technology	Advantages	Challenges	References
MEC-assisted AD	• High energy recovery. • High methane production and yield. • Diverse bacterial communities. • High DIET efficiency. • Disintegration of complex organic compounds.	• High capital cost. • High operation cost. • Nature of electrodes' material. • Conductivity of electrodes.	[7,46,59,62,63,68,78,203]
IntensiCarb	• Complete decoupling of HRT from SRT. • High potential for wide range of applications. • High nutrients separation and recovery. • Enhancement in sludge thickening.	• Novel technology that has multiple scientific and technological unknowns. • Lack of knowledge of the effect of vacuum on the involved microbes.	[186,188,189]
MHP using C. bescii	• Effective in degrading the recalcitrant wastes (e.g., manure and dairy wastes). • Producing thermostable enzymes.	• High production cost for the feedstock. • Low hydrogen yield per mass of substrate in some cases. • High capital and operation cost. • Sensitivity of *C.bescii* to operational conditions.	[193,197,198]
Hydrogen injection	• High biogas production rate. • Less capital cost. • Less footprint requirements for the treatment facility. • High and diverse microbial activity. • Efficient DIET and interspecies exchange.	• pH increase and regulation. • Residual hydrogen at high hydrogen injection rates. • Increase in the hydrogen partial pressure. • High hydrogen cost.	[171,172,175–178,180,181,185]
Micro-aeration	• High solubilization of substrates. • Diverse bacterial and archaeal communities. • Better conductive environment and DIET. • Cost-efficient and environment-friendly technology.	• Automatic process control. • Management of automatic online data acquisition schemes. • Measuring and testing of multiple physicochemical parameters online.	[2,118,119]
Conductive functional materials	• Low cost. • Less energy requirements. • High mechanical strength properties. • Enhancing DIET pathway. • Formation of electro-active biofilm formation. • Reduced redox potential.	• Potential toxic effect on microbes. • Not effective for complex substrates. • Adsorption of organics at high additive concentrations. • Poor mass transfer because of relatively small surface area in some big-sized materials (e.g., carbon cloth). • High possibility of being lost in the discharge for particle-sized materials (e.g., GAC). • Complexity in operation and retrieval for particle-sized materials	[90–97,105,113,114]
AnMBRs	• Decoupling of HRT and SRT. • High resistance against toxicity. • High biogas production. • Suitable for industrial wastewater treatment. • Simultaneous organics and nutrient removal. • Reduction in greenhouse gases emissions.	• Poor performance in harsh conditions. • High capital cost. • Membrane fouling challenges. • Difficulties in determining the optimal bioreactor design and operation scheme.	[137,163–167,170]

integration of thermophilic MEC-AD technology and bioaugmentation enhanced the biogas production rate by 7 % compared to the non-bioaugmented reactor [70]. Qin et al. (2021) found that the incorporation of MEC-AD and magnetite enhanced the methane yield by 9.4 % while the abundance of methanogens was increased by 1.2 time compared to a conventional MEC-AD system [67]. In a lab-scale study, addition of magnetite to a MEC-AD enhanced the removal of COD by 95 % [208]. Yang et al. (2022) reported that the addition of Fenton-like reagents to a MEC-AD increased methane production (up to 63.8 %) and abundance of hydrogenotrophic methanogens (61.16 %) [209].

The addition of conductive functional materials in AD process was integrated with different technologies to enhance the biogas production rate, solubilization of complex organics and removal of VSS (Fig. 13). In multiple studies in the literature, it was emphasized that the conductive functional materials can be easily integrated with nano-zero-valent iron to enhance the AD processes [92,210,211]. Activated carbon was combined with trypsin and nano-zero-valent iron to enhance the methane production in AD of WAS, resulting in higher activity of electroactive and methanogens [212]. In addition, GAC and granular biochar (GBC) were combined with thermophilic pretreatment to enhance the digestion of multiple wastes such as WAS, blackwater, wheat husk, kitchen waste, food waste and sewage sludge [41,98,213,214]. Carbon conductive additives can also be combined with alkaline pretreatment to allow better disintegration of complex compounds with high methane yield and production rate [215]. Moreover, the combination between conductive materials and MEC-AD systems maximized the biogas production rate and microbial activity in anaerobic digesters [85,208].

Micro-aeration technology can be combined with other intensification and pretreatment techniques to enhance the hydrolysis process, solubilization of particulate compounds and production of biogas (Fig. 13). Montalvo et al. (2018) found that the pretreatment process of mixed sewage sludge using micro-aeration and metallic wastes (e.g., copper mining residues and fly ash) significantly enhanced the methane yield where the methane production increased by 162 – 200 % compared to untreated control reactors [120]. In addition, the micro-aeration technology can be integrated with thermophilic pretreatment (at 55 °C or higher), resulting in an effective digestion process (i.e., thermophilic microaerobic pretreatment) since it accelerates the hydrolysis reactions with efficient organic matter utilization and volatile fatty acids production [131,132]. The micro-aeration technique can be also combined with ferric chloride pretreatment for efficient digestion, yielding in enhanced methanogenesis reactions through efficient hydrogenotrophic pathways [216]. The micro-aeration technology can be also integrated with GAC to enhance the reduction of VSS and total COD [217]. The conductive materials can be employed with micro-aeration technology to maximize the removal of VSS and COD with increasing the methane production rate [133].

Novel integrations for the abovementioned technologies can be applied to overcome some of their challenges and improve the AD efficiency as well. Membrane fouling is the main challenge for AnMBR digesting sludge to be widely used in full-scale applications [135]. Integration of MEC and AnMBR can potentially reduce the membrane fouling and improve the AD process by using a conductive membrane as a cathode. The electrostatic repulsion between the membrane surface and organics (negative charge) can be enhanced, leading to elongation of membrane fouling rate. Besides, the electric field will enhance the

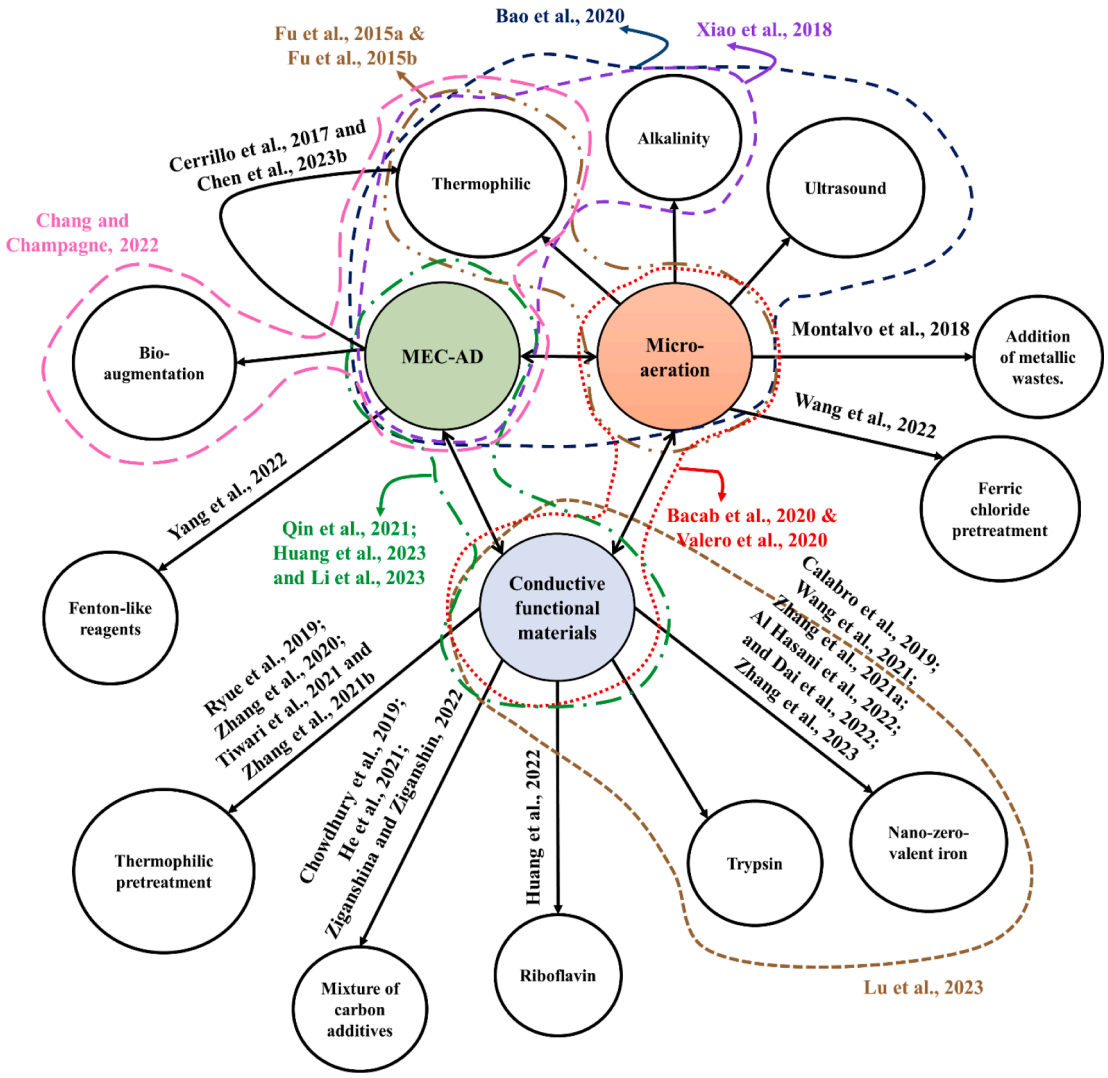


Fig. 13. Integration of innovative technologies with other AD pretreatment and intensification techniques.

activity of the microorganisms, leading to higher methane production rate [218].

Hydrogen injection in anaerobic digester is coupled with pH increase due to depletion of CO_2, causing an inhibition to the methanogens. Therefore, it is important to control the pH at the optimal range, which can be achieved by co-digestion with acidic waste or removal of ammonia. Luo and Angelidaki (2013) controlled the pH lower than 8.0 using acidic whey as a co-substrate in a digester fed with manure during hydrogen injection, which did not lead to inhibition of the AD process [182]. Integration of hydrogen assisted anaerobic digesters and IntensiCarb can be a practical option to control the pH and increase ammonia recovery. The pH increase due to hydrogen injection will result in conversion of a fraction of ionized ammonia to the non-ionized (gaseous) ammonia, which can be easily stripped in the thermal vacuum unit. Besides, recovery of ammonia would result in pH decrease to the optimal range. Thus, more stable conditions for the AD process can be achieved and ammonia recovery can be maximized.

5. Technology readiness level

Technology readiness level (TRL) is a framework/method to assess the maturity of a certain technology from basic principles to commercialization levels, allowing a consistent comparative evaluation of various technologies across industries [219]. The TRL can be used as an indication to determine the required resources and capabilities for developing technologies at different scales [220]. The TRL of a technology is represented on a scale from one to nine, where TRL of nine corresponds to the most mature technology (i.e., commercial scale) while TRL of one indicates the basic theoretical form of the technology [221,222]. The TRL scale of 1 – 3 refers to lab scale while TRL scale of 4 – 6 and 7 – 9 are defined as pilot and commercial scales, respectively [223]. The TRL scale was selected in the current study due to its wide application and promising outcomes in assessing the maturity of emerging technologies in both industrial and academic contexts. In the current review, the main evaluation criteria of the TRL scale included the technical maturity and scalability of the AD innovative intensification technologies. In other words, the extent of applying these technologies in lab-, pilot-, and full-scale systems was the main driving criteria of determining the TRL scale of each technology. The assessment of the TRL scale of each technology was evaluated according to the outcomes of the comprehensive analysis and investigations in the current review (Section 2). Further details and information about the TRL scale can be found in previous publications in the literature that were oriented more towards the principles and characteristics of TRL [219–223].

Table 4 presents the TRL and its description for each technology based on this review. For MEC-AD, most of the research focused on development of the process on the lab-scale, confirming the increase in

Table 4
TRL of the innovative intensification technologies.

Technology	TRL	Description
Micro-aeration AD	7	System model or prototype demonstration in operational environment
AnMBR	7	System model or prototype demonstration in operational environment
MEC-AD	5	Technology validated in relevant environment
Conductive functional materials	5	Technology validated in relevant environment
Hydrogen injection	5	Technology validated in relevant environment
IntensiCarb	4	Technology validated in lab
MHP using C. bescii	4	Technology validated in lab

methane production and yield and showing its economic benefits. However, long-term operation of MEC-AD treating WAS at larger scale showed that the actual energy efficiency was not more than 50 % of the theoretical energy efficiency [224]. Therefore, there is still a need for further investigations to improve the process efficiency, which indicates that MEC-AD technology lies in TRL of five. Similarly, conductive functional materials were extensively used in lab-scale system with limited studies at larger scale, thus, using these materials in AD sits around TRL of five. Integration of micro-aeration with AD has largely proven successful implementation on lab- and pilot-scale systems, which showed a reduction in H_2S content and an increase in methane yield. However, there is still a need for further research and development to convert the outcomes obtained from the pilot-scale implementation into full-scale applications, in which automatic process control of micro-aeration was one of the main challenges [2]. Therefore, TRL of micro-aeration technology for AD can be quantified to be seven. Similarly, use of AnMBR for sludge digestion has been investigated on lab- and pilot-scale systems with very limited attempts on full-scale level, which was mainly due to difficulties of membrane fouling control [135]. As results, the TRL of the AnMBR technology can be quantified to be seven.

Hydrogen injection in anaerobic digester has proven to be a practical option to convert CO_2 into methane. Most of the research were conducted on lab-scale experiments. Few studies have investigated the hydrogen injection in a full-scale digester, where hydrogen utilization was much lower than lab-scale systems with controlled conditions [225]. Thus, the TRL of hydrogen injection in anaerobic digester is five. For IntensiCarb, all of the studies have been conducted on lab-scale analyses, showing the process intensification with no adverse effects on biogas production [189]. Therefore, there is a need to scale-up this process to verify the lab-scale findings. The IntensiCarb technology sits at TRL of four. Application of *C. bescii* in a pretreatment step prior to the AD process was mainly investigated in the laboratory, where it was found that *C. bescii* increased conversion of waste into biogas by double or more. However, more research is needed to optimize the retention times of this process [192]. Therefore, the MHP using C. bescii is still at the TRL of four.

6. Conclusions

The AD process plays a crucial role in sludge management in wastewater treatment plants. It converts the organic matter into methane-rich biogas, which is used to recover energy. The AD process is highly sensitive to some operational conditions, such as temperature and pH, with relatively slow biochemical reaction rates, especially in the in the hydrolysis and methanogenesis stages. In the current study, multiple intensification technologies for enhanced AD processes have been reviewed, including MEC-AD, micro-aeration, hydrogen injection, conductive functional materials, AnMBR, IntensiCarb, and MHP using *C. bescii*. All technologies showed a substantial improvement in biogas production and solids reduction. However, each technology has some limitations that hinder its implementation in full-scale AD applications, which requires further investigations. Among these technologies, IntensiCarb and MHP using *C. bescii* seem to have the fewest limitations. However, these technologies have been mainly investigated on the lab scale, which need further research to scale up these processes to verify the findings reported in the literature. Different integrations can be applied to overcome some of the limitations of each technology, such as the integration of MEC and AnMBR to reduce membrane fouling and improve the AD process using a conductive membrane as a cathode. Therefore, more research is needed to overcome the limitations reported for each technology and implement these technologies on a full-scale. In general, the main mechanisms, governing factors, efficiency gains, advantages, challenges, integration with other technologies, and technology readiness level were examined for each of these seven technologies. The main research outcomes garnered from this review can serve as a valuable reference for identifying potential intensification strategies that can be seamlessly applied in AD applications pertaining to biogas upgrading while recommending the next steps for overcoming the challenges associated with these technologies.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.enconman.2024.118979.

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